```
import pandas as pd
```

```
df = pd.read_csv("customer_shopping_data.csv")
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 99457 entries, 0 to 99456
Data columns (total 10 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   invoice_no      99457 non-null  object
 1   customer_id     99457 non-null  object
 2   gender          99457 non-null  object
 3   age             99457 non-null  int64
 4   category        99457 non-null  object
 5   quantity        99457 non-null  int64
 6   price           99457 non-null  float64
 7   payment_method  99457 non-null  object
 8   invoice_date    99457 non-null  object
 9   shopping_mall   99457 non-null  object
dtypes: float64(1), int64(2), object(7)
memory usage: 7.6+ MB
```

```
print(df.isna().sum().sum())
```

```
0
```

```
df["invoice_date"] = pd.to_datetime(df["invoice_date"], dayfirst=True)
```

```
print(df.duplicated().sum())
```

```
0
```

```
print(df["invoice_no"].duplicated().sum())
```

```
0
```

```
print(df[["age", "quantity", "price"]].describe())
```

```
                age      quantity         price
count  99457.000000  99457.000000  99457.000000
mean      43.427089      3.003429    689.256321
std       14.990054      1.413025    941.184567
min       18.000000      1.000000      5.230000
25%       30.000000      2.000000     45.450000
50%       43.000000      3.000000    203.300000
75%       56.000000      4.000000   1200.320000
max       69.000000      5.000000   5250.000000
```

```
df["year"] = df["invoice_date"].dt.year
df["month"] = df["invoice_date"].dt.month
df["month_name"] = df["invoice_date"].dt.month_name()
df["week_day"] = df["invoice_date"].dt.day_name()
df["day"] = df["invoice_date"].dt.day
df["age_group"] = pd.cut(df["age"], bins=[17, 25, 35, 45, 55, 65, 100],
                         labels=["18-25", "26-35", "36-45", "46-55", "56-65", "65+"])
```

```
df.head()
```

	invoice_no	customer_id	gender	age	category	quantity	price	payment_method	invoice_date	shopping_mall	year	mont
0	I138884	C241288	Female	28	Clothing	5	1500.40	Credit Card	2022-08-05	Kanyon	2022	
1	I317333	C111565	Male	21	Shoes	3	1800.51	Debit Card	2021-12-12	Forum Istanbul	2021	1
2	I127801	C266599	Male	20	Clothing	1	300.08	Cash	2021-11-09	Metrocity	2021	1
3	I173702	C988172	Female	66	Shoes	5	3000.85	Credit Card	2021-05-16	Metropol AVM	2021	
4	I337046	C189076	Female	53	Books	4	60.60	Cash	2021-10-24	Kanyon	2021	1

Pasos siguientes: Generar código con df | New interactive sheet

```
cols_texto = ["invoice_no", "customer_id", "gender", "category", "payment_method", "shopping_mall"]
for col in cols_texto:
```

```
    print(col, "->", (df[col] != df[col].str.strip()).sum())
```

```
invoice_no -> 0
customer_id -> 0
gender -> 0
category -> 0
payment_method -> 0
shopping_mall -> 0
```

```
df.to_csv("customer_shopping_clean.csv", index=False)
```

```
from google.colab import files
files.download("customer_shopping_clean.csv")
```

Empieza a programar o a crear código con IA.

```
print(df['price'].sum())
```

```
68551365.91
```

```
df["age"].min()
```

```
18
```

```
df["age"].max()
```

```
69
```

Empieza a programar o a crear código con IA.